

SIMCOE, ON, Canada

WMO: 715274

Lat: 42.850N

Lon: 80.270W

Elev: 241

StdP: 98.47

Time Zone: -5.00 (NAE)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.2	-15.0	-22.3	0.5	-16.4	-19.7	0.7	-14.5	9.7	-0.4	8.8	-2.9	2.3	270	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	30.0	23.2	28.3	21.9	27.0	21.1	24.7	28.3	23.5	26.8	22.3	25.4	3.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	19.0	26.7	22.4	17.6	25.5	21.3	16.4	24.1	76.1	28.3	71.2	26.8	66.7	25.5	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.3	6.4	-21.1	32.5	2.5	2.3	-22.9	34.2	-24.4	35.6	-25.8	36.9	-27.6	38.6		
(5)			-21.1	26.0	3.1	0.9	-23.3	26.7	-25.1	27.2	-26.8	27.7	-29.0	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-4.0	-4.1	0.3	6.8	13.3	18.3	21.1	20.0	15.3	9.6	3.4	-1.7		
	DBStd	10.12	5.85	5.49	5.92	4.62	4.46	3.78	3.11	3.13	4.08	4.33	4.65	5.36		
	HDD10.0	1906	434	396	306	118	16	1	0	0	5	61	205	364		
	HDD18.3	3914	692	629	558	347	168	46	8	18	107	271	448	622		
	CDD10.0	1272	0	0	7	22	118	250	343	311	164	50	7	1		
	CDD18.3	240	0	0	0	2	12	46	93	70	17	1	0	0		
	CDH23.3	1705	0	0	1	13	107	351	714	450	67	3	0	0		
	CDH26.7	389	0	0	0	3	15	83	199	83	6	0	0	0		
(14) Wind	WSAvg	2.6	3.4	3.0	3.1	3.0	2.5	2.2	1.8	1.6	2.0	2.4	3.1	2.8		
(15) Precipitation	PrecAvg	984	69	58	86	92	75	87	76	79	93	74	85	99		
	PrecMax	1189	140	131	169	140	136	161	154	273	235	172	165	165		
	PrecMin	739	16	9	16	47	34	34	32	31	9	16	32	35		
	PrecStd	106	33	32	43	30	27	32	32	54	57	38	35	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	11.1	21.8	25.4	28.5	31.2	33.6	30.7	27.3	23.5	17.8	13.0		
		MCWB	10.7	8.0	15.9	17.5	20.0	22.6	22.9	24.7	21.5	18.0	14.0	11.9		
	2%	DB	9.0	7.7	16.9	19.7	26.3	29.0	30.6	28.7	25.1	20.7	15.4	10.3		
		MCWB	8.1	5.2	12.0	13.8	19.1	21.4	23.8	23.2	20.4	16.0	13.1	8.8		
	5%	DB	5.5	4.7	12.5	16.9	24.1	27.0	28.7	27.2	23.6	18.7	13.4	6.9		
		MCWB	4.4	3.0	9.6	11.9	17.6	19.8	22.4	22.0	19.2	15.0	11.7	5.3		
	10%	DB	3.2	2.8	8.5	14.4	21.6	25.2	27.1	25.9	22.1	16.8	10.9	4.5		
		MCWB	2.2	1.7	5.8	10.0	16.3	19.2	21.6	21.2	18.4	13.8	9.1	3.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	8.8	16.2	17.7	21.9	25.0	26.5	26.0	23.1	19.0	15.4	12.8		
		MCDB	11.9	9.8	21.2	24.1	25.5	29.5	28.9	28.7	25.9	22.0	16.7	12.8		
	2%	WB	8.2	5.2	12.7	15.2	20.5	22.7	24.9	24.6	21.4	17.4	13.7	8.8		
		MCDB	8.6	7.3	15.3	18.7	24.4	27.2	28.5	27.8	23.7	19.6	14.8	9.7		
	5%	WB	4.7	3.3	9.6	12.7	18.6	21.4	23.7	23.3	20.4	15.9	11.8	5.6		
		MCDB	5.4	4.5	13.0	15.7	23.4	25.2	27.4	26.0	22.6	18.0	13.6	6.7		
	10%	WB	2.3	1.7	6.1	10.5	16.6	20.2	22.5	22.1	19.3	14.3	9.2	3.4		
		MCDB	3.1	2.7	8.3	13.7	20.9	23.7	25.9	24.7	21.6	16.4	10.8	4.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	7.4	8.1	9.4	10.4	10.3	10.1	9.8	9.7	8.9	6.7	6.0		
		MCDBR	8.1	8.8	13.5	13.1	13.4	12.3	11.5	10.5	10.2	10.8	9.3	8.2		
	5% WB	MCWBR	7.8	7.5	9.7	8.6	7.9	6.9	5.9	6.1	6.9	7.5	8.0	7.6		
		MCWBR	8.2	8.4	13.1	11.8	12.3	10.0	10.5	9.6	9.4	9.3	9.5	8.2		
(40) Clear-Sky Solar Irradiance	taub		0.293	0.343	0.334	0.373	0.394	0.431	0.424	0.409	0.377	0.357	0.311	0.291		
		taud	2.403	2.206	2.368	2.320	2.310	2.239	2.280	2.307	2.375	2.415	2.512	2.476		
	Ebn at Noon		847	841	908	893	880	845	847	847	851	823	824	820		
		Edh at Noon	77	111	107	122	127	137	130	122	106	90	70	66		
(44) All-Sky Solar Radiation	RadAvg		1.45	2.13	3.37	4.49	5.53	6.12	6.09	5.34	4.22	2.57	1.59	1.00		
	RadStd		0.13	0.26	0.40	0.48	0.53	0.41	0.42	0.28	0.39	0.24	0.23	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page